

Thursday 2 August 2018
2.00 pm – 3.00 pm
(Duration: 1 hour)

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

INTERNET TECHNOLOGY (M)

This examination paper is worth a total of 60 marks.

Section A (30 marks) consists of 10 multiple choice questions.

Section B (30 marks) comprise 13 open questions.

Answer all questions in each section

The use of a calculator is not permitted in this examination

INSTRUCTIONS TO INVIGILATORS

**Please collect all exam question papers and exam
answer scripts and retain for school to collect.
Candidates must not remove exam question papers.**

Section A

30 marks

This section consists of 10 multiple-choice questions. A correct answer is worth 3 marks, an incorrect answer will result in a deduction of 1 mark (i.e., -1 marks), whilst no answer will result in 0 marks. Only one answer is correct.

The following form contained in a Django template applies to questions 1 and 2.

```
<form action="{% url 'polls:vote' question.id %}"
      method="post">
  {% csrf_token %}
  {% for choice in question.choice_set.all %}
    <input type="radio" name="choice_var" id="choice
      {{ forloop.counter }}" value="{{ choice.id }}" />
    <label for="choice{{ forloop.counter }}">
      {{ choice.choice_text }}</label><br />
  {% endfor %}
  <input type="submit" value="Vote" />
</form>
```

1. What is the main purpose of `{% csrf_token %}` in the above template?
 - A. to protect and secure the HTTP POST action that is initiated on the subsequent submission of a form
 - B. to let Django middleware know that a form has been submitted
 - C. to request a confirmation token from the server
 - D. to initiate a response/request protocol
2. From the above Django template, when the form is submitted, an HTTP POST request will occur and a key-value pair will be transmitted as part of the message. Which of the following statements is true?
 - A. The key is “choice” and the value is an integer.
 - B. The key is “choice_var” and the value is given by choice_text.
 - C. The key is “choice” and the value is “choice” concatenated with an integer.
 - D. The key is “choice_var” and the value is an integer.
3. Which line of code selects the element with a class called **dogs** and assigns an event to when the element is double clicked?
 - A. `$(".dogs").ondblclick( )`
 - B. `$("#dogs").dblclick( )`
 - C. `$("#dogs").ondblclick( )`
 - D. `$(".dogs").dblclick( )`

4. Consider the following XML document:

```
<employees>
  <employee>
    <firstName>John</firstName> <lastName>Doe</lastName>
  </employee>
  <employee>
    <firstName>Anna</firstName> <lastName>Smith</lastName>
  </employee>
</employees>
```

Which one of the following options gives the equivalent JSON representation?

- A. `[{"employees":[{"firstName":"John", "lastName":"Doe"}, {"firstName":"Anna", "lastName":"Smith"}]}`
 - B. `{"employees":[{"firstName":"John", "lastName":"Doe"}, {"firstName":"Anna", "lastName":"Smith"}]}`
 - C. `{"employees":["{"firstName":"John", "lastName":"Doe"}, {"firstName":"Anna", "lastName":"Smith"}]}`
 - D. `{"employees":{"{"firstName":"John", "lastName":"Doe"}, {"firstName":"Anna", "lastName":"Smith"}}}`
5. Most web application frameworks make use of the Model View Controller Template (MVCT) architecture (or some variant of this). Which one of the following options best describes how Django implements the MVCT architecture?
- A. Models are implemented using SQL commands that act directly on the database tables; Views are responsible for handling and processing specific requests; the Controller specifies which view should be called by pattern-matching a URL in `views.py`; Templates are responsible for rendering the data to be presented.
 - B. Models are implemented using SQL commands that act directly on the database tables; Views are responsible for handling and processing specific requests; the Controller specifies which view should be called by pattern-matching a URL in `urls.py`; Templates that render the data to be presented are contained in `views.py`.
 - C. Models are implemented by an Object-Relational Mapping to actual database tables; Views are responsible for handling and processing specific requests; the Controller specifies which view should be called by pattern-matching a URL in `views.py`; Templates are responsible for rendering the data to be presented.
 - D. Models are implemented by an Object-Relational Mapping to actual database tables; Views are responsible for handling and processing specific requests; the Controller specifies which view should be called by pattern-matching a URL in `urls.py`; Templates are responsible for rendering the data to be presented.

6. Consider the following HTML, JavaScript and JSON code:

HTML
<code><div id="placeholder"></div></code>
JavaScript
<pre>var data={"users":[{ "firstName":"Speedy", "lastName":"Gonzales", "joined":2010 }, { "firstName":" Lightning", "lastName":"Rodriguez", "joined":2013 }]}</pre> <pre>document.getElementById("placeholder").innerHTML=data.users[1].firstName + " " + data.users[0].lastName + " " + data.users[1].joined;</pre>

How would the text in the client browser render?

- A. Speedy Gonzales 2010
- B. Lightning Gonzales 2013
- C. Speedy Rodriguez 2010
- D. Lightning Rodriguez 2013

7. Consider the following HTML code:

```
<body>
  <div style="background-color: yellow">Internet Technology</div>
</body>
```

The CSS referencing is an example of which type of referencing?

- A. inline referencing
- B. embedded referencing
- C. external referencing
- D. online referencing

8. Developing web applications typically involves developing distributed applications that run on web browsers. What are the problems and benefits associated with developing applications in such an environment?
- A. Difficult to tamper with the application, requires session management, no access to local file system, and high barrier to installation.
 - B. Control over the client application version, flexible user interface, low entry barrier to installation, and application can be released often.
 - C. Control over client application version, easy to update the application, performance issues, and needs to handle a variety of browser/devices.
 - D. Low entry barrier to installation, application can be released often, requires session management, and there is no application container model.
9. DOM Parsing and SAX Parsing are two methods of parsing XML documents. Which of the following characteristics best describes DOM and SAX Parsing?
- A. DOM is appropriate for using parts of the data, especially if there are memory constraints, whilst SAX is appropriate if you need the whole document.
 - B. DOM builds a hierarchical model of XML elements in memory, whilst SAX provides an event-driven parser.
 - C. DOM provides an event-driven parser, whilst SAX is appropriate if you need to move about the elements of the document freely.
 - D. DOM is appropriate for using parts of the data in the order they appear in the file, whilst SAX is appropriate if you need to move about the elements of the document freely.
10. Which of the following deliverables are the responsibilities of an Information Architect? Select the answer that best applies.
- A. user personas, site maps and URL designs
 - B. user personas, wireframes and templates
 - C. site maps, URL designs and Cascading Stylesheets
 - D. system architecture diagrams, user needs matrices and usability testing

Section B

30 marks

A restaurant conglomerate would like to create a site to allow their customers to rate hamburgers eaten. Basically, the conglomerate asked your team to design, specify and develop a “Rate My Burger” application with the following minimum requirements:

- This site should allow registered customers to enter reviews, providing a rating and comments.
- The site should allow registered members of the conglomerate to add a new restaurant, where the member has to enter the restaurant's name, description, photo, address, city and postcode.
- For each new restaurant, the exact address of a location is provided through a web service provider (e.g. Google Maps).
- Burgers at each restaurant have a number of descriptors, which include the restaurant, the place where it can be eaten, numeric ratings (ratings from 1 to 10; 10 being the best), comments, and a photo.
- The site should let customers browse the most liked burgers and be able to see all the comments and ratings.
- The front page should display the latest added burgers and comments, the best burger of the month and show which burgers are near to the customer’s location. Each burger is sold at only one restaurant whilst a restaurant can sell many burgers.

11. Draw the high-level system architecture for this application. Describe what each component is responsible for.

[5]

12. Draw the Entity Relationship (E-R) Diagram for this application using compressed Chen notation. Do not include the attributes in your E-R diagram.

[10]

13. Assume that you have created **models.py** and you have declared the restaurant location model (class name: **Location**). How do you define the relationship between the restaurant location and burger in Django? Hint: The model should be stored in a **restaurant** variable within the **Burger** class model.

[2]

14. Assume that in **views.py**, you have created an **add_burger** view. How do you define the URL mapping (in your **urls.py**) to the **add_burger** view using URL defined with a named URL pattern “**add_burger**” and with “**add_burger/**” URL? Hint: Use URL pattern matching

[1]

Now someone in your team has created the following template which is intended to allow users to add a new burger (burger_form.html) but your team member needs your help to complete the following:

```
{% if reviews %}
<ul>
    <!-- See question 15 -- >                                (R1)
    <div class="review">
        {{comment.text}}
        <br>
        <div class="profile_img">
            {% if comment.user.picture %}
                 (R2)
            {% else %}
                
            {% endif %}
        </div>
        <div class="review_right">
            {{comment.user}}<br>
            {{comment.date}}
            <!-- See question 17 -- >                            (R3)
        {% endfor %}
    </ul>
{% else %}
    <strong>No review currently in {{burger.name}}.</strong><br>
{% endif %}
```

15. What do you need to insert at point R1 such that the for loop is complete? [1]
16. What do you need to insert at point R2 in order to display the user's image correctly? [1]
17. What do you need to insert at point R3 such that the template is a valid HTML file? [1]

The newest member of your team started coding the “BurgerCategories” model; however, this member is having problems with Django’s slug field and slugify functions as below:

```
class BurgerCategories(models.Model):
    name = models.CharField(max_length=128, unique=True)
    # See question 18                                (R4)

    def save(self, *args, **kwargs):
        # See question 19                            (R5)
        super(BurgerCategories, self).save(*args, **kwargs)

    def __unicode__(self):
        return self.name
```

18. Can you help your colleague to define a **unique category slug field** at point R4 of the code? [1]

19. What do you need to include at point R5 of the code in order to create a slug for the category's name? [1]

Your team leader has now asked you to write the code required to handle the index request in **views.py** as below, but another member started working on the same task and there's now a merge conflict in the company's Git repository:

```
def index(request):
    best= Comments.objects.extra(select={'sum':'rating'}).order_by('-sum')
    # See question 20 (R6)
    result = {}
    if best:
        result['best'] = best[0]
    if comments:
        result['reviews'] = comments

    return # See question 21 (R7)
```

20. Can you rewrite the code at point R6? In your Comments model, you have an attribute **date** defined as DateField and you want to retrieve a list of comments ordered in descending order in terms of the date. [2]

21. What do you need to return at point R7 in order to render correctly your index page? Your template name should be 'burger/index.html' [1]

22. In the app's admin.py, how do you register the **Restaurant/Location** model such that it can be viewed in the **admin site**? Your models are defined in **burger.models**; *remember to import what it is needed!* [2]

23. You have finished your work for today and you need to update the team's repository with your latest updates. What are the commands you need to use (as you type them in a terminal/command prompt) in order to update your team's remote Git repository with your new code? [2]